

Heart Rate & Oxygen Sensor (MAX30102)

Hardware Overview

What is it?

The **MAX30102** is a high-sensitivity optical sensor that measures Heart Rate (BPM) and Blood Oxygen Saturation (SpO2). It is found in smartwatches and hospital pulse oximeters.

Electrical Explanation

The sensor uses two LEDs and a sensitive light detector (photodiode) to "look" through skin:

- Red LED: Used primarily to calculate Oxygen Saturation.
- Infrared (IR) LED: Used for the Heart Rate and identifying if a finger is present.

How the Math Works

The library runs a "Math Engine" in the background to process the data.

- Heart Rate: Every time your heart beats, a "wave" of blood rushes into your finger. This blood absorbs more Infrared (IR) light, causing a drop in the light bouncing back. The library measures the time between waves to calculate beats per minute.
- Oxygen Saturation (SpO2): Hemoglobin absorbs Red and Infrared light differently based on its oxygen level. The library calculates the ratio between these signals and uses a formula to provide a percentage, usually 95% - 100% for healthy users.

Key Hardware Facts

- **Connection**: I²C Bus (SDA/SCL).
- **Default Address**: 0x57.
- **Logic Voltage**: 3.3V or 5V
- **Signal Flow**: LEDs shine light into skin → Light bounces back → Photodiode measures intensity → ADC converts to digital numbers → I²C → Processor calculates vitals.

Software & Programming

kcomp_AHT10.h

To get accurate results, the sensor needs to be read very fast without stopping.

Usage: `#include <kcomp_MAX30102.h>`

Function Reference

Function Name	Description & Parameters
<code>bool initMAX30102 ()</code>	Setup: Initializes the sensor and LEDs. Must be called in <code>setup ()</code> .
<code>void updateVitalSigns ()</code>	The Worker: CRITICAL! Call this in your <code>loop ()</code> as often as possible. If you don't call this, the math stops working.
<code>int getHeartRate ()</code>	Read BPM: Returns your heart rate. Returns 0 if no finger is detected.
<code>int getSPO2 ()</code>	Read Oxygen: Returns the percentage of oxygen in your blood.

⚠ Pitfalls & Troubleshooting

- Movement and Noise

The sensor measures tiny changes in light. Shaking your finger will create "noise".

The Fix: Keep your finger very still and apply gentle and steady with low pressure. Pressing too hard will cut off blood flow and stop the reading.

- Blocking Code

When using `delay(1000)` or `long ledFade` functions, the `updateVitalSigns()` function cannot run. This will cause the vital signs to become frozen or incorrect.

The Fix: Use Non-Blocking timers (like the `millis()` -method shown).



Sample Program

This program initializes the sensor and prints your vitals every second.

```
#include <kcomp_MAX30102.h>

// Variable to keep track of the last time
// data was printed to the serial monitor
unsigned long last_print = 0;

void setup() {
    // Initialize the serial communication at 115200 baud rate
    Serial.begin(115200);
    // Initialize the MAX30102 sensor. Halt the program if it fails.
    if (!initMAX30102()) {
        Serial.println("Sensor Error!");
        while(1);
    }
    Serial.println("Place finger on sensor...");
}

void loop() {
    // Continually update the vital signs using the sensor data
    updateVitalSigns();

    // Print the data to the serial monitor once every second
    if (millis() - last_print > 1000) {
        // Update the time of the last print operation
        last_print = millis();

        // Get the current heart rate and SpO2 values
        int bpm = getHeartRate();
        int spo2 = getSPO2();

        // If a valid heart rate is detected, print the BPM and SpO2 values
        if (bpm > 0) {
            Serial.print("BPM: "); Serial.print(bpm);
            Serial.print(" | SpO2: "); Serial.print(spo2);
            Serial.println("%");
        } else {
            // If no valid heart rate is detected, prompt the user
            Serial.println("Reading... (Keep Still)");
        }
    }
}
```

💡 **Try it out:** You can find the full sample program **HeartRateOxygenMAX30102** in the K-Comp examples folder!